Assignment 4

1. Write a Relational Algebra expression to perform **Cartesian Product (Cross Join)** between Emp and Dept tables.

```
mysql> select * from Emp cross join Dept;
+-------+---------+----------+--------+-----------+--------+-----------+----
------+
| EmpID | EmpName | Salary   | DeptID | ManagerID | DeptID | DeptName  | Loc
ation |
+-------+---------+----------+--------+-----------+--------+-----------+----
------+
|   101 | Amit    | 60000.00 |     10 |      NULL |     50 | Sales     | Del
hi    |
|   101 | Amit    | 60000.00 |     10 |      NULL |     40 | Marketing | Che
nnai  |
|   101 | Amit    | 60000.00 |     10 |      NULL |     30 | Finance   | Mum
bai   |
|   101 | Amit    | 60000.00 |     10 |      NULL |     20 | HR        | Pun
e     |
|   101 | Amit    | 60000.00 |     10 |      NULL |     10 | IT        | Mum
bai   |
|   102 | Priya   | 75000.00 |     10 |       101 |     50 | Sales     | Del
hi    |
|   102 | Priya   | 75000.00 |     10 |       101 |     40 | Marketing | Che
nnai  |
|   102 | Priya   | 75000.00 |     10 |       101 |     30 | Finance   | Mum
bai   |
|   102 | Priya   | 75000.00 |     10 |       101 |     20 | HR        | Pun
e     |
|   102 | Priya   | 75000.00 |     10 |       101 |     10 | IT        | Mum
bai   |
|   103 | Rahul   | 45000.00 |     20 |       101 |     50 | Sales     | Del
hi    |
|   103 | Rahul   | 45000.00 |     20 |       101 |     40 | Marketing | Che
nnai  |
|   103 | Rahul   | 45000.00 |     20 |       101 |     30 | Finance   | Mum
bai   |
|   103 | Rahul   | 45000.00 |     20 |       101 |     20 | HR        | Pun
e     |
|   103 | Rahul   | 45000.00 |     20 |       101 |     10 | IT        | Mum
bai   |
|   104 | Sneha   | 90000.00 |     30 |       102 |     50 | Sales     | Del
hi    |
|   104 | Sneha   | 90000.00 |     30 |       102 |     40 | Marketing | Che
nnai  |
|   104 | Sneha   | 90000.00 |     30 |       102 |     30 | Finance   | Mum
```

2.Write a SQL query to perform an **INNER JOIN** between Emp and Dept tables displaying EmpName and DeptName.

```
mysql> select e.EmpName, d.DeptName from Emp e
    -> inner join Dept d
    -> on e.DeptID = d.DeptID;
+---------+-----------+
| EmpName | DeptName  |
+---------+-----------+
| Amit    | IT        |
| Priya   | IT        |
| Rahul   | HR        |
| Sneha   | Finance   |
| Neha    | Finance   |
|  s      | Marketing |
+---------+-----------+
6 rows in set (0.02 sec)
```

3. Write a SQL query to perform a **LEFT OUTER JOIN** between Emp and Dept tables to display all employees along with their department information, including employees without a department.

```
mysql> select e.EmpName,d.DeptName from Emp e
    -> Left Join Dept d
    -> on e.DeptId = d.DeptId;
+---------+-----------+
| EmpName | DeptName  |
+---------+-----------+
| Amit    | IT        |
| Priya   | IT        |
| Rahul   | HR        |
| Sneha   | Finance   |
| Vikas   | Marketing |
| Neha    | Finance   |
|  an     | NULL      |
+---------+-----------+
7 rows in set (0.01 sec)
```

4. Write a SQL query to perform a **RIGHT OUTER JOIN** between Emp and Dept tables.

```
mysql> select e.EmpName,d.DeptName from Emp e
    -> Right Join Dept d
    -> on e.DeptId= d.DeptId;
+---------+-----------+
| EmpName | DeptName  |
+---------+-----------+
| Amit    | IT        |
| Priya   | IT        |
| Rahul   | HR        |
| Sneha   | Finance   |
| Neha    | Finance   |
| Vikas   | Marketing |
|         | Sales     |
+---------+-----------+
7 rows in set (0.00 sec)
```

5. Write a SQL query to perform a **NATURAL JOIN** between Emp and Dept.

```
mysql> select * from Emp
    -> Natural Join Dept;
+--------+-------+---------+----------+-----------+-----------+----------+
| DeptID | EmpID | EmpName | Salary   | ManagerID | DeptName  | Location |
+--------+-------+---------+----------+-----------+-----------+----------+
|     10 |   101 | Amit    | 60000.00 |      NULL | IT        | Mumbai   |
|     10 |   102 | Priya   | 75000.00 |       101 | IT        | Mumbai   |
|     20 |   103 | Rahul   | 45000.00 |       101 | HR        | Pune     |
|     30 |   104 | Sneha   | 90000.00 |       102 | Finance   | Mumbai   |
|     30 |   106 | Neha    | 80000.00 |       101 | Finance   | Mumbai   |
|     40 |   105 | Vikas   | 50000.00 |       102 | Marketing | Chennai  |
+--------+-------+---------+----------+-----------+-----------+----------+
6 rows in set (0.05 sec)
```

6. Write a SQL query to perform a **CROSS JOIN** between Emp and Project table.

```
mysql> SELECT * FROM Emp
    -> CROSS JOIN Project;
+-------+---------+----------+--------+-----------+-----------+-------------
--------+-------+-----------+
| EmpID | EmpName | Salary   | DeptID | ManagerID | ProjectID | ProjectName
        | EmpID | Budget    |
+-------+---------+----------+--------+-----------+-----------+-------------
--------+-------+-----------+
|   101 | Amit    | 60000.00 |     10 |      NULL |         5 | Analytics Da
shboard |   106 | 450000.00 |
|   101 | Amit    | 60000.00 |     10 |      NULL |         4 | Sales Portal
        |   105 | 250000.00 |
|   101 | Amit    | 60000.00 |     10 |      NULL |         3 | Library Mana
gement  |   103 | 200000.00 |
|   101 | Amit    | 60000.00 |     10 |      NULL |         2 | Payroll Syst
em      |   102 | 300000.00 |
|   101 | Amit    | 60000.00 |     10 |      NULL |         1 | Banking Syst
em      |   101 | 500000.00 |
|   102 | Priya   | 75000.00 |     10 |       101 |         5 | Analytics Da
shboard |   106 | 450000.00 |
|   102 | Priya   | 75000.00 |     10 |       101 |         4 | Sales Portal
        |   105 | 250000.00 |
|   102 | Priya   | 75000.00 |     10 |       101 |         3 | Library Mana
gement  |   103 | 200000.00 |
|   102 | Priya   | 75000.00 |     10 |       101 |         2 | Payroll Syst
em      |   102 | 300000.00 |
|   102 | Priya   | 75000.00 |     10 |       101 |         1 | Banking Syst
em      |   101 | 500000.00 |
|   103 | Rahul   | 45000.00 |     20 |       101 |         5 | Analytics Da
shboard |   106 | 450000.00 |
|   103 | Rahul   | 45000.00 |     20 |       101 |         4 | Sales Portal
        |   105 | 250000.00 |
```

7.Write a SQL query to **create a new table EmpBackup with the same structure as Emp but no data**.

```
mysql> CREATE TABLE EmpBackup LIKE EMP;
Query OK, 0 rows affected (0.15 sec)

mysql> DESC EMPBACKUP;
+-----------+--------------+------+-----+---------+-------+
| Field     | Type         | Null | Key | Default | Extra |
+-----------+--------------+------+-----+---------+-------+
| EmpID     | int          | NO   | PRI | NULL    |       |
| EmpName   | varchar(50)  | NO   |     | NULL    |       |
| Salary    | decimal(10,2)| YES  |     | NULL    |       |
| DeptID    | int          | YES  | MUL | NULL    |       |
| ManagerID | int          | YES  | MUL | NULL    |       |
+-----------+--------------+------+-----+---------+-------+
5 rows in set (0.04 sec)
```

8. Write a SQL query to **copy all data from Emp into EmpBackup table**.

```
mysql> Select * from EmpBackup;
+-------+---------+----------+--------+-----------+
| EmpID | EmpName | Salary   | DeptID | ManagerID |
+-------+---------+----------+--------+-----------+
|   101 | Amit    | 60000.00 |     10 |      NULL |
|   102 | Priya   | 75000.00 |     10 |       101 |
|   103 | Rahul   | 45000.00 |     20 |       101 |
|   104 | Sneha   | 90000.00 |     30 |       102 |
|   105 | Vikas   | 50000.00 |     40 |       102 |
|   106 | Neha    | 80000.00 |     30 |       101 |
|   107 | Karan   | 40000.00 |   NULL |       101 |
+-------+---------+----------+--------+-----------+
7 rows in set (0.00 sec)
```

9. Write a SQL query to **create a new table ProjectArchive with the same structure and data as Project**.

```
mysql> CREATE TABLE ProjectArchive AS
    ->
    ->
    -> SELECT * FROM  CProjec^C
mysql> create table ProjectArchive As
    -> select * from Project;
Query OK, 5 rows affected (0.09 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> select * from ProjectArchive;
+-----------+--------------------+-------+-----------+
| ProjectID | ProjectName        | EmpID | Budget    |
+-----------+--------------------+-------+-----------+
|         1 | Banking System     |   101 | 500000.00 |
|         2 | Payroll System     |   102 | 300000.00 |
|         3 | Library Management |   103 | 200000.00 |
|         4 | Sales Portal       |   105 | 250000.00 |
|         5 | Analytics Dashboard |   106 | 450000.00 |
+-----------+--------------------+-------+-----------+
5 rows in set (0.00 sec)
```

11. Write a SQL query using a **subquery** to find all employees whose salary is greater than the **average salary of all employees**.

```
mysql> SELECT * FROM Emp
    -> WHERE SALARY > (
    -> SELECT AVG(Salary)
    -> FROM Emp);
+-------+---------+----------+--------+-----------+
| EmpID | EmpName | Salary   | DeptID | ManagerID |
+-------+---------+----------+--------+-----------+
|   102 | Priya   | 75000.00 |     10 |       101 |
|   104 | Sneha   | 90000.00 |     30 |       102 |
|   106 | Neha    | 80000.00 |     30 |       101 |
+-------+---------+----------+--------+-----------+
3 rows in set (0.03 sec)
```

12.Write a SQL query using a **subquery in WHERE clause** to find employees working in departments located in 'Mumbai'.

```
mysql> select * from emp where deptId in(select deptid from dept where locat
ion= 'mumbai');
+-------+---------+----------+--------+-----------+
| EmpID | EmpName | Salary   | DeptID | ManagerID |
+-------+---------+----------+--------+-----------+
|   101 | Amit    | 60000.00 |     10 |      NULL |
|   102 | Priya   | 75000.00 |     10 |       101 |
|   104 | Sneha   | 90000.00 |     30 |       102 |
|   106 | Neha    | 80000.00 |     30 |       101 |
+-------+---------+----------+--------+-----------+
4 rows in set (0.02 sec)
```

Step 6: Why use `IN` instead of `=`?

This is an important concept, bro!

- `=` is suitable when the subquery returns a single value.
- `IN` is suitable when the subquery can return multiple values.

Here, Mumbai has two departments, so `IN` is the right choice.

13. Write a SQL query to display all departments where the number of employees is greater than **5**, using a subquery.

```
mysql> SELECT *
    -> FROM Dept
    -> WHERE DeptID IN (
    ->     SELECT DeptID
    ->     FROM Emp
    ->     WHERE DeptID IS NOT NULL
    ->     GROUP BY DeptID
    ->     HAVING COUNT(*) > 1
    -> );
+--------+----------+----------+
| DeptID | DeptName | Location |
+--------+----------+----------+
|     10 | IT       | Mumbai   |
|     30 | Finance  | Mumbai   |
+--------+----------+----------+
2 rows in set (0.00 sec)
```

14. Write a subqueries for find out Max salary of employee.

```
mysql> SELECT *
    -> FROM Emp
    -> WHERE Salary = (
    ->     SELECT MAX(Salary)
    ->     FROM Emp
    -> );
+-------+---------+----------+--------+-----------+
| EmpID | EmpName | Salary   | DeptID | ManagerID |
+-------+---------+----------+--------+-----------+
|   104 | Sneha   | 90000.00 |     30 |       102 |
+-------+---------+----------+--------+-----------+
1 row in set (0.05 sec)
```

. 15 Write a self join query for each table.

```
mysql> SELECT
    ->     E.EmpName AS Employee,
    ->     M.EmpName AS Manager
    -> FROM Emp E
    -> LEFT JOIN Emp M
    -> ON E.ManagerID = M.EmpID;
+----------+---------+
| Employee | Manager |
+----------+---------+
| Amit     | NULL    |
| Priya    | Amit    |
| Rahul    | Amit    |
| Sneha    | Priya   |
| Vikas    | Priya   |
| Neha     | Amit    |
| [illegible]n | Amit |
+----------+---------+
7 rows in set (0.01 sec)
```